

# Computing GC Content using functions and conditionals

---

BILD 62

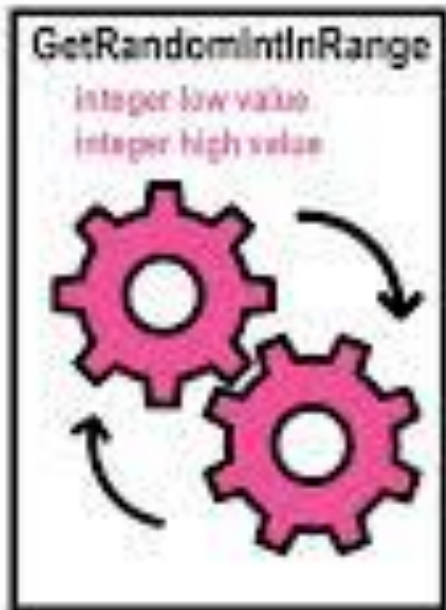
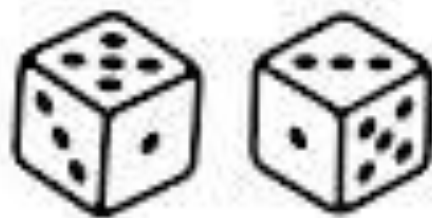
By the end of  
this lecture you  
will be able to:

- Recognize function syntax & write a simple function
  - Use functions to start writing more modular code in Python
-

Functions are pieces of code that are designed to do *one* *task*

Functions take in inputs, process those inputs, and *possibly* return an output.

Python has *built-in* functions, but we can also write our own!



# function syntax

function  
name

def function(value) :

print(value)

function  
body

indented  
by 4 spaces  
(or tab)

colon

The diagram illustrates the syntax of a Python function definition. It shows the code `def function(value) :` on the first line and `print(value)` on the second line, which is indented. Annotations include: a green arrow pointing from 'function name' to 'function' in the first line; a red arrow pointing from 'colon' to the colon at the end of the first line; a red arrow pointing from 'indented by 4 spaces (or tab)' to the indentation of the second line; and a large right-facing curly bracket on the right side of the second line, with 'function body' written next to it.

# function syntax

**input arguments** (these can be variables or default arguments)

```
def function(b):
```

```
    a = b**2
```

```
    return a
```

**return** to retrieve a variable outside of a function (*what happens in the function stays in the function*)  
**ALSO ENDS THE FUNCTION!**

**call to function** giving it the argument and saving the returned variable as a

```
a = function(6)
```

# function syntax

```
def function(b):
```

```
    a = b**2
```

```
    return a
```

**defines function** (creates it so that we can use)

```
a = function(6)
```

**call to function** giving it the argument and saving the returned variable as a

# function syntax

```
def function(b):
```

```
    c = b**2
```

```
    a = c * 2
```

```
    return a
```

```
a = function(6)
```

```
print(c)
```

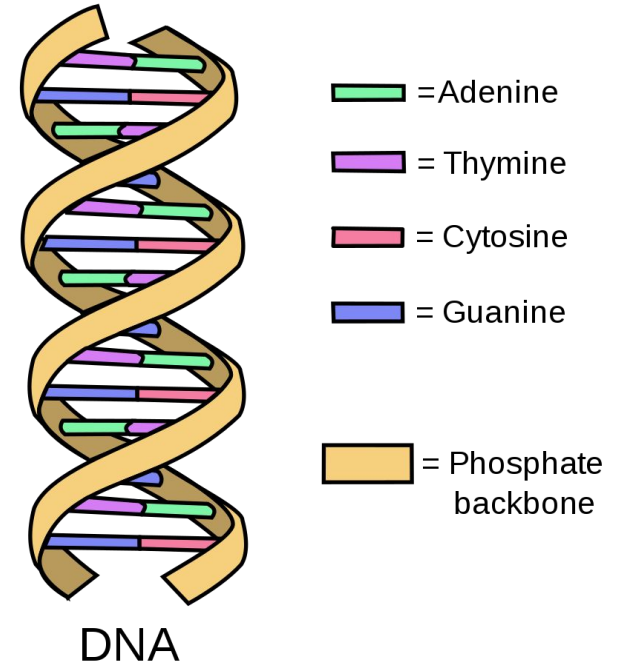
????

**return** to retrieve a variable outside  
of a function (*what happens in the  
function stays in the function*)



# DNA Refresher

- Nucleic acids contain all of the information to build our cells!
- In deoxyribonucleic acid (DNA) there are four different: **adenine (A)**, **cytosine (C)**, **guanine (G)**, and **thymine (T)**.
- The sequence of a nucleic acid polymer is defined by the order of these bases, which we can represent with a string of A's, C's, G's, and T's.
- **Base pairs:** A bonds to T, and C bonds to G



# Representing DNA on a computer

5' - ATTCGTCA - 3'

Forward strand

3' - TAAGCAGT - 5'

Reverse strand

} **same # of G or C,**  
so we can work  
with either strand

One way to characterize & distinguish different sequences of DNA is by their **GC content**. Can we write a **program** that does this?

# Resources

[Python Tutorial: Functions](#)

Calling Functions in Python: <https://www.youtube.com/watch?v=8ja0Pz8CJ5o>